WELCOME

Beyond Chatbots; Deep into AI

Our Purpose & Reason for Visit

Who We Are

We are second year students from SCU currently doing our IT degree

Our Purpose

We are going to give you an insight into what AI is and how it can be integrated through out life

OUR TEAM

Meet the Members

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WORKSHOP SESSION

AI Fundamentals

Core Concepts

Daily Applications

Advancements

Future Directions

What is AI?

AI simulates human intelligence through systems that learn, reason, and decide autonomously.

01

CORE CONCEPTS

Machine Learning

Systems that learn from data without explicit programming

Neural Networks

Mimic brain patterns for image recognition & more

Transformers

Enable large language models via attention mechanisms

NLP & Vision

Text analysis and visual processing powering chatbots & self-driving cars

02 Daily Applications

Health Tracking

Smartwatches detect heart irregularities in real time

Smart Shopping

Apps like Amazon predict your next reorder automatically

Productivity

AI summarises emails and schedules your day for you

Smart Home

IoT devices adjust lights & temperature based on your habits

Personalization

YouTube & Netflix learn your taste from viewing patterns

Navigation & More

Google Maps, virtual assistants, and health monitoring apps

03 Recent Advancements

GPT-5.4 · March 2026

Next-Gen Language Models

- 1-million-token context window
- Excels at complex desktop tasks
- Outperforms humans on OSWorld-V benchmark

Hardware Innovation

Neuromorphic Chips

- Mimic biological brain structure
- Solve physics equations efficiently
- Aid climate modelling with low energy use

How LLMs Work

Key LLM Concepts

- Tokenization: breaking text into chunks
- Embeddings: meaning as numbers
- Agents: AI that takes actions autonomously

04 Future Directions

Predictive Analytics

AI anticipates trends, risks, and opportunities before they happen

Explainable AI

Transparent systems that show their reasoning — critical for trust

Advanced Robotics

Autonomous robots for assembly lines, surgery, and space exploration

Human-AI Collaboration

AI as a creative partner in design, research, and decision-making

Smart Cities & IoT

Interconnected AI managing traffic, energy, and urban infrastructure

Energy-Efficient Hardware

Next-gen chips making powerful AI available everywhere, cheaply

KEY TAKEAWAYS

Remember these

1. AI learns from data. It's not magic, it's mathematics
2. You're already using AI every day without realising it
3. GPT-5.4 and neuromorphic chips are redefining what's possible right now
4. The future belongs to those who learn to collaborate with AI

LIVE BUILD

Creating a webpage for the SCU LEO club

BRIEF

You are building a webpage for the SCU Leo Club.

The Leo Club is a student-led community at SLIIT City Uni focused on leadership, volunteering, and making a difference on campus and beyond.

YOUR PAGE MUST INCLUDE:

1. A compelling hero section with the club's name and a tagline
2. A short 'About Us' section — what the club does and why it matters
3. A highlights section — 3 reasons to join (e.g. leadership, community, events)
4. A call-to-action — a 'Join Now' or 'Get Involved' button

JUDGED ON:

- Does it make someone want to join?
- Is it clear, attractive, and well-structured?
- Does it feel like it belongs to SCU?

You have 20 minutes. Use any AI tool.

Can you give us a prompt?

PROMPT

```
Build a modern, professional landing page for SCU (SLIIT City Uni).

PURPOSE: Attract prospective students and showcase the university's strengths.

SECTIONS TO INCLUDE:
1. Hero section — university name, a compelling tagline, and a call-to-action button
(e.g. 'Apply Now')
2. About Us — 2-3 sentences on the university's mission and values
3. Why Choose Us — 3 to 4 key strengths displayed as cards (e.g. experienced teachers,
modern facilities, strong community)
4. Courses Offered — a clean grid showing subject areas or streams available
5. Testimonials — 2 short quotes from fictional students
6. Contact / Footer — address, phone number, email, and social media links

DESIGN REQUIREMENTS:
- Color scheme: deep blue and white with gold accents
- Clean, modern layout with clear visual hierarchy
- Mobile-friendly and professional in appearance
- Use bold headings and short, punchy copy throughout
```

New Trends in Job Fields

How AI Is Reshaping the Future of Work

Based on Anthropic's 2026 Labor Market Impact Research

AI Exposure

Job Trends

Coding with AI

THE BIG QUESTION

Is AI taking our jobs? or changing them?

Past Fear

~25% of US jobs deemed 'offshorable' in 2009.
A decade later? Most maintained healthy growth.

AI Reality

AI adoption is real but gradual. Actual coverage is far below theoretical capability.

The Truth

AI is shifting which skills matter — not eliminating work entirely.

How Do We Measure AI Exposure?

01 O*NET Database

~800 US occupations broken into individual tasks

02 Anthropic Usage Data

Real Claude conversations categorized by task type & work context

03 Eloundou et al. (2023)

Theoretical LLM capability score (β) per task can AI speed it up 2×?

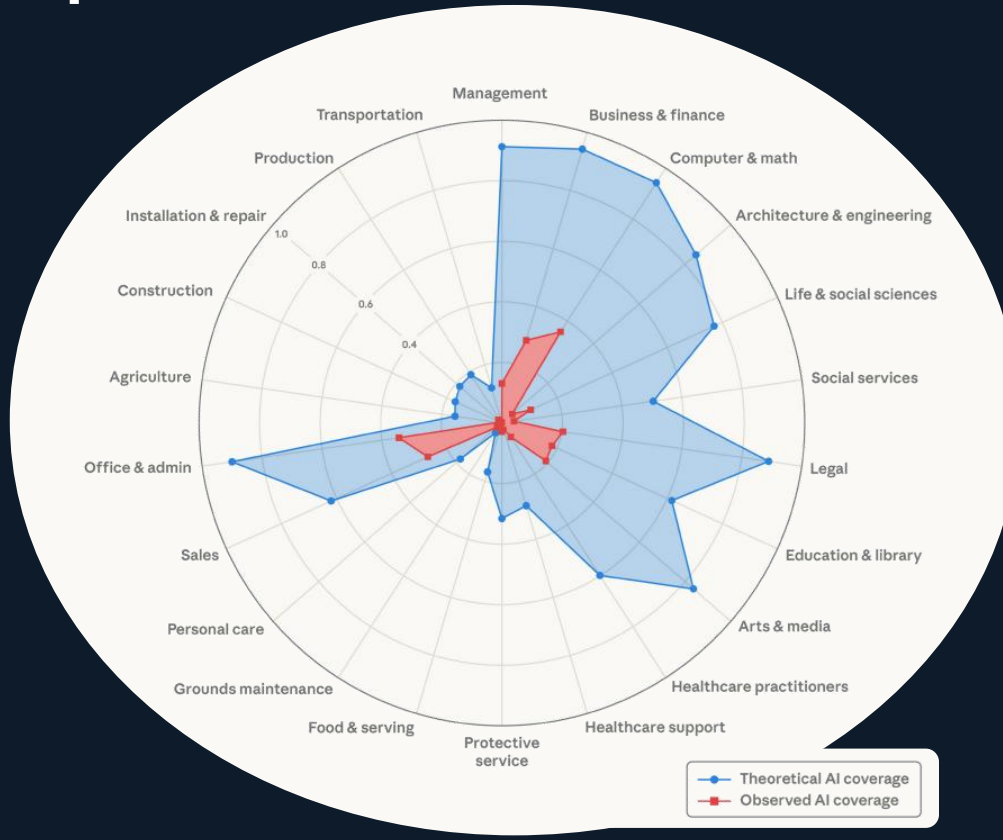
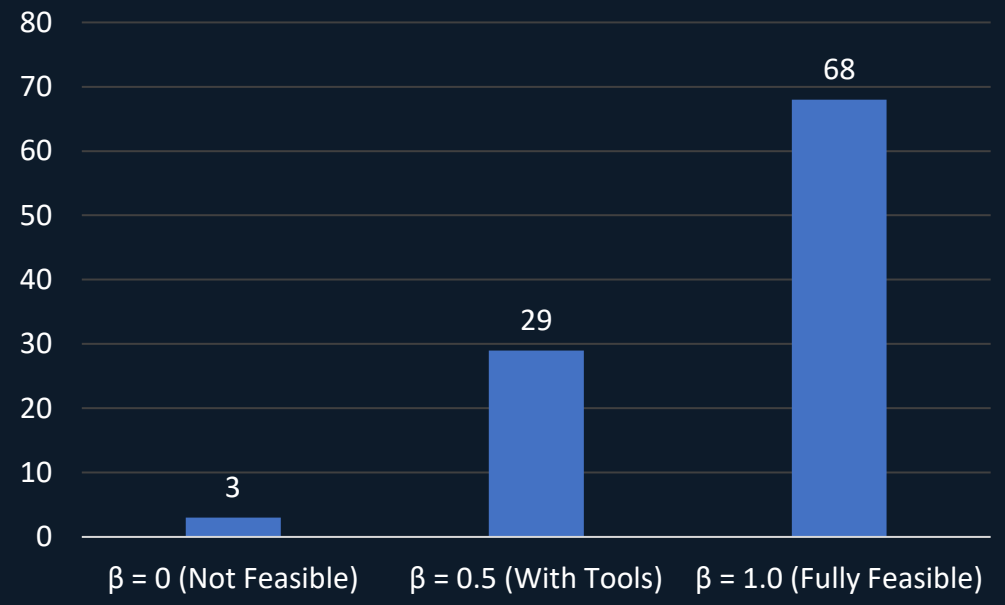


FIGURE 1 | THEORY VS. REALITY

97% of Claude Tasks Fall in AI-Feasible Categories



68% Fully feasible for LLMs alone

29% Feasible with additional tools

3% Not feasible

Source: Anthropic Economic Index (Aug/Nov 2025 data)

FIGURE 2 | BY OCCUPATIONAL CATEGORY

AI Is Far From Reaching Its Theoretical Potential

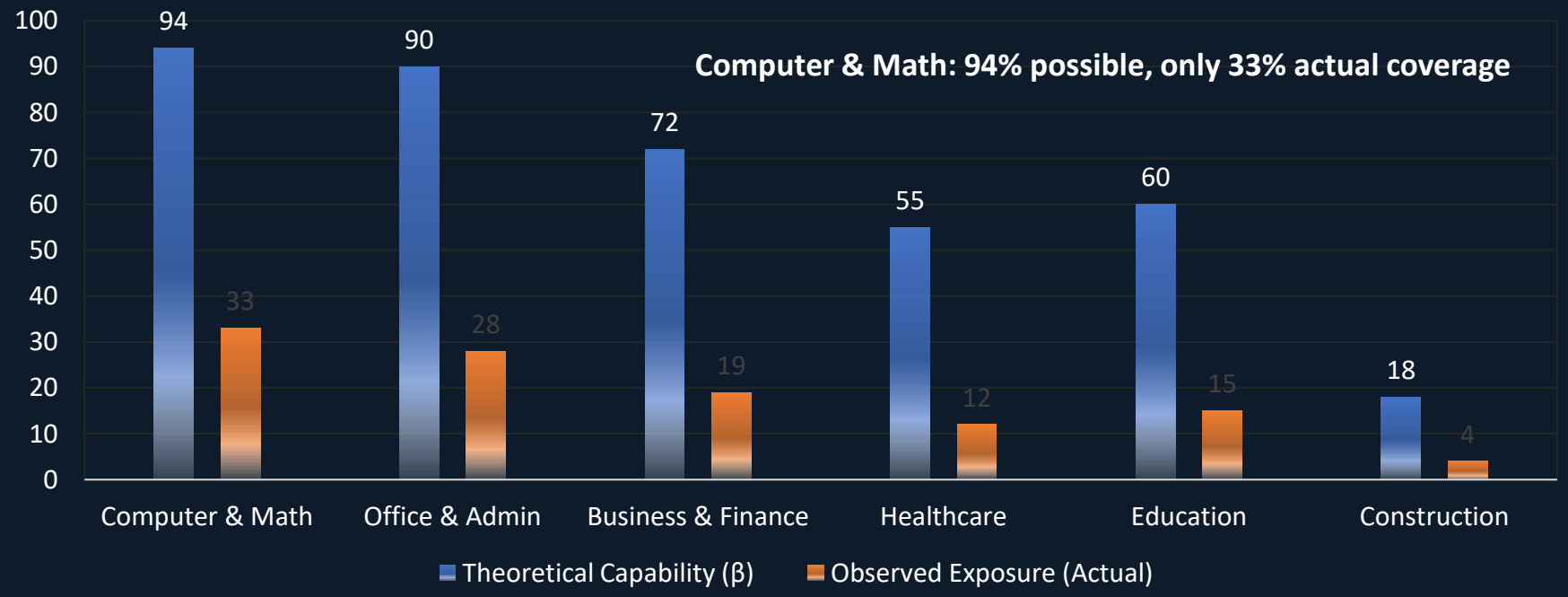


FIGURE 3 | MOST EXPOSED OCCUPATIONS

Top 10 Most AI-Exposed Job Roles

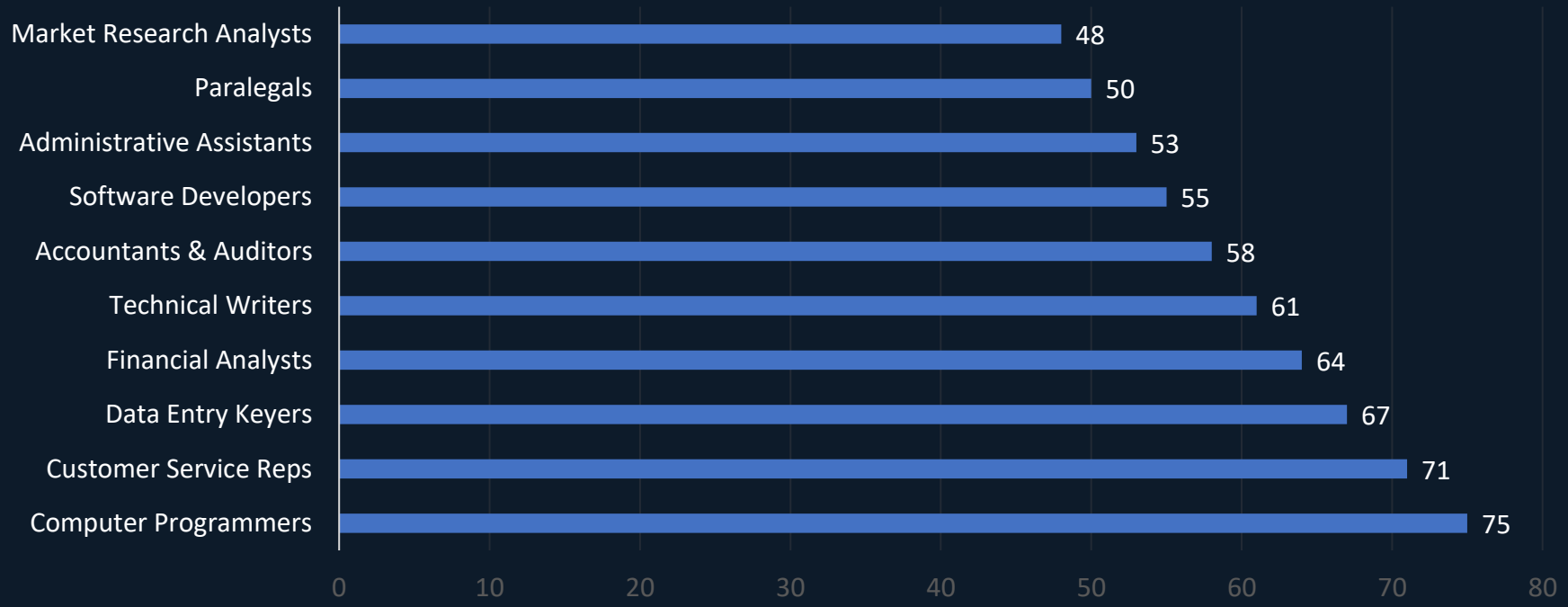


FIGURE 5 | WORKER CHARACTERISTICS

Who Is Most Exposed to AI?

Comparing top-quartile exposure workers vs. zero-exposure workers (Pre-ChatGPT, Aug–Oct 2022)

+16%
More likely to be female

+47%
Higher salary on average

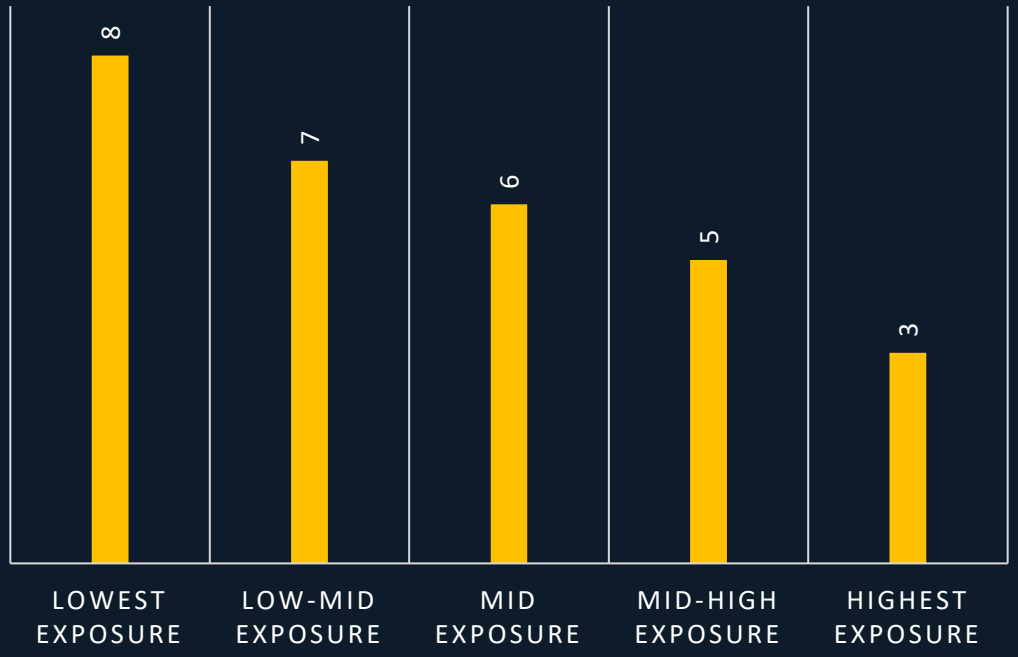
+11%
More likely to be white

4.5×
More likely to have grad degree

⚠ **AI primarily affects educated, higher-paid, office-based workers, not low-wage physical workers**

FIGURE 4 | PROJECTED JOB GROWTH 2024–2034

More AI Exposure = Lower Job Growth Projections



–0.6%

drop in job growth for every 10% increase in AI exposure

Computer Programmers	-3.2%	75% exposure
Healthcare Aides	+22%	8% exposure
Cooks & Food Workers	+10%	0% exposure

No Unemployment Spike — But Hiring Is Slowing for Young Workers

Overall Unemployment: No Change

No significant increase in unemployment for highly exposed workers (post-ChatGPT)

The difference between exposed vs. non-exposed workers is statistically indistinguishable from zero

A '2× unemployment' scenario for top exposed workers would still be detectable and we don't see it

Young Workers (22–25): Hiring Slows

–14%

drop in job-finding rate for young workers entering exposed occupations

⚠ Young workers may be staying in existing jobs, switching fields, or returning to school instead

⚠ Separate study (Brynjolfsson et al.) finds 6–16% drop in employment for exposed occupations, age 22–25

KEY TAKEAWAY | CAREER STRATEGY

How to Position Yourself in the AI Era

Become the AI Operator

Use AI tools as force multipliers. A developer using Claude Code or Cursor can do 10× the work of one who doesn't.

Specialize in Judgment

AI handles repetitive tasks well. Complex judgment, ethics, client relationships, creativity — still human territory.

T-Shaped Skills

Deep expertise in one area + broad AI literacy. The combination is extremely rare and highly valuable right now.

Stay in Exposed Fields

Don't avoid AI-exposed jobs — learn to thrive in them. Programmers using AI are MORE in demand, not less.

Coding with AI Tools & Agents

Claude Code • OpenClaw • Cursor • Aider • Windsurf

From vibe-coding to full agentic pipelines — what works, what doesn't, what's next

The Spectrum: Autocomplete → Autonomous Agents

Autocomplete	Chat-in-IDE	Pair Programmer	Agentic Coder	Multi-step Agent	Fully Auto Pipeline

← *Less Autonomous*

More Autonomous →

GitHub Copilot
Autocomplete + Chat

Cursor
AI-native IDE

Claude Code
Agentic terminal CLI

OpenClaw (Cline)
Open-source agent

Aider
Git-aware agent

CLAUDE CODE

Anthropic's Agentic Coding CLI

Terminal-Native

Runs directly in your terminal. No IDE required. Uses your existing tools: git, npm, pip, bash.

Full Codebase Context

Reads your entire repo structure. Understands relationships between files.

Multi-step Execution

Plans, writes code, runs tests, fixes bugs — all in one shot. Not just suggestions.

Safe by Design

Ask permission before running commands. Shows diffs before writing. Audit every action.

```
$ npm install -g @anthropic-ai/claude-code   |   then: $ claude
```

CURSOR | AI-NATIVE IDE

Cursor: The AI-First Code Editor

Tab Completion	Predicts multi-line completions in context. Way beyond single-token autocomplete.
⌘K Inline Edit	Select code → describe what to change. Applies the edit directly in-place.
⌘L Chat Panel	Chat with your codebase. Ask 'why does this break' and get an answer with file context.
Composer (Agent)	Multi-file edits. 'Refactor auth to use JWT' — it plans and does it across 10 files.
.cursorrules	Custom instructions for your project. AI knows your stack, coding standards, conventions.

Cursor vs. VS Code + Copilot

Feature	Copilot	Cursor
Multi-file edit	Limited	**Full**
Codebase Q&A	Partial	**Deep**
Agent mode	Basic	**Advanced**
.rules config	No	**Yes**
Model choice	Fixed	**Any LLM**

Open-Source AI Coding Agents You Should Know

Cline / OpenClaw

VS Code Extension • Open Source • Free

→ Runs as VS Code extension — brings agent features to your existing editor

→ Use any model: Claude, GPT-4o, Gemini, local LLMs via Ollama

→ Full file system access: reads, writes, creates files autonomously

→ Executes terminal commands (with your permission)

→ Supports MCP servers for custom tool integrations

→ OpenClaw = community fork with extra features + UI improvements

Aider

Terminal-based • Git-native • Benchmark leader

→ Works entirely in the terminal — ideal for server environments

→ Deeply git-integrated: auto-commits after each AI edit

→ One of the highest SWE-Bench scores among open tools

→ Repo map feature: AI understands the whole codebase structure

→ Voice coding mode with whisper integration

→ 'aider --model claude-3-7-sonnet-20250219' — use any Claude model

Which Tool Should You Use?

Tool	Best For	Interface	Cost	Skill Level
Claude Code	End-to-end task automation, repos	Terminal CLI	$20/mo Pro	Intermediate
Cursor	Everyday coding, IDE workflow	IDE (VS Code fork)	$20/mo	Beginner+
Cline / OpenClaw	Flexible model choice, open-source	VS Code Extension	Free (pay per token)	Intermediate

Which Tool Should You Use?

Tool	Best For	Interface	Cost	Skill Level
Aider	Git projects, no GUI needed	Terminal CLI	Free (pay per token)	Advanced
Copilot	Quick autocomplete, simple Q&A	Any IDE plugin	$10/mo	Beginner
Windsurf	Deep codebase indexing, flows	IDE (VS Code fork)	$15/mo	Intermediate

How to Get Great Code from AI

Prompt Quality Makes All the Difference

❌ Vague Prompt

```
"Make a login system"
```

✅ Specific Prompt

```
"Express.js JWT login. bcrypt passwords.
Return {token, userId} or 401. Include
input validation."
```

Give Context — Tell the AI your stack, constraints, existing code style

Be Explicit — Name the library, framework, and expected behavior

One Task at a Time — Break big features into steps — AI works best in focused chunks

Review Everything — AI code can have bugs, outdated patterns, security gaps — always review

The Agentic Loop

What Can Agents Use as Tools?

- Read/Write Files
- Run Terminal Commands
- Search the Web
- Call APIs
- Execute Code
- Read Git History

Key Takeaways

1 **AI is already reshaping hiring**

Computer programmers, analysts, and admin roles have 50 - 75% task exposure.

2 **No mass unemployment yet**

But young workers are entering exposed fields at lower rates, the signal is early.

3 **The best coders use AI**

Claude Code, Cursor, Cline; these are professional tools, not shortcuts.

4 **Prompting is a skill**

The quality of your output depends on the quality of your input. Practice it.

5 **Adapt or get left behind**

Every field is seeing AI exposure. Are you the operator or the one being replaced?

anthropic.com/research/labor-market-impacts • claude.ai/code • cursor.sh • github.com/cline/cline • aider.chat

A Big Question asked by most of the people

Q: Is my code private when I use these tools?

It depends on the tool and plan. Anthropic's Claude Code on claude.ai does not use your conversations to train models when you're a paid subscriber. Cursor has a privacy mode that disables sending code to their servers. Always check the data policy before using these tools on proprietary codebases. When in doubt, use local models via Ollama with Cline/OpenClaw no data leaves your machine.

COMMON QUESTIONS

Q: Which tool should I start with as a beginner?

Cursor. It's the lowest friction entry point if you already use VS Code, it's a drop-in replacement. The UI is familiar, and features like Tab completion and Command-K inline edits work immediately without any configuration.

Q: Will using AI make me a worse programmer?

Only if you stop reading and understanding the code it produces. The risk is cargo-culting blindly accepting AI output without knowing why it works. The antidote is reviewing everything, asking the AI to explain its decisions, and treating AI-generated code the same way you'd treat a Stack Overflow answer: a starting point, not gospel.

COMMON QUESTIONS

Q: Are these tools useful for non-coders?

Increasingly, yes. Claude Code can help with data analysis scripts, automating repetitive file tasks, building simple internal tools, and writing automation workflows even for people who don't consider themselves developers. The barrier to entry for 'good enough' code has dropped dramatically.

Q: What's the difference between Claude Code and just using Claude.ai for coding?

Claude.ai chat is conversational you paste code in, get answers back, copy-paste manually. Claude Code is agentic it lives in your terminal, reads your actual files, runs commands, and makes changes directly to your codebase. Claude.ai is like asking a smart friend for advice. Claude Code is like having that friend sit down at your computer and work alongside you.

Can you give us a prompt?

PROMPT

Create a professional advertisement image for SCU (SLIIT City Uni).

CONCEPT: An inspiring, aspirational scene that makes prospective students want to be part of this university.

IMAGE DETAILS:
- Setting: A bright, modern university campus with lush greenery
- Subjects: A diverse group of enthusiastic university students, smiling and engaged
- Mood: Energetic, hopeful, and ambitious — a place where futures are built
- Lighting: Golden hour sunlight, warm and cinematic
- Composition: Wide shot with students in the foreground, university building visible in the background

TEXT OVERLAY (if supported):
- Headline: 'Your Future Starts Here'
- Sub-text: 'SCU (SLIIT City Uni) — Shaping Tomorrow's Leaders'

STYLE:
- Photorealistic, high resolution, suitable for social media and print
- Color palette: warm golds, deep blues, and bright whites
- Clean and modern — no clutter, strong visual focus on the students

NOW IT'S YOUR TURN

Thank You

for having us.